

Discrete Structures: Evidence Before Claims

Sept. 3 recap — GPU data reasoning and container setup

COMSC 2043

The lesson

A clean exit code is not proof of anything. Inspect the evidence before you trust the claim.

Cold start vs. warm start

- Not about temperature — about whether the GPU already has a job queued.
- A warmed-up pipeline finishes the next job faster than the same hardware starting cold.
- Same idea as a car engine, applied to model calls on a GPU.

From observation to modest claim

- 1 Notice something in the data (e.g. the newest GPU wasn't the fastest).
- 2 That is an observation, not yet a claim.
- 3 A **modest claim** is no bigger than what the evidence actually supports.
- 4 Ask: what would it take to convince a skeptic?

Sampling bias, live

- Six machines, one building, is not a random sample of “all GPUs.”
- Same failure mode as the WWII “armor the bullet holes” story.
- You can only reason about the evidence you actually have.

Hitting a real wall, live

- Podman/WSL2 setup verification failed live, unstaged, on the instructor's own machine.
- Correct response to an administrative/platform wall: stop, file a help desk ticket, move on.
- Not every failure is yours to fix with more elbow grease.

The AI assistant got it wrong, live

- Asked to find a folder while already standing inside it.
- Like using your phone as a flashlight to find your phone.
- The skill is noticing the mismatch, not typing the first prompt.

Two-pass builds and Git hygiene

- LaTeX's first pass writes supporting files (e.g. a table of contents).
- The second pass reads that back in and renders it correctly.
- One pass with no visible change is normal, not a failure.
- `.gitignore` the build byproduct (`main.pdf`); track the source.

Your next move

Write one observation from the GPU Showdown table as a single, modest, falsifiable claim. Bring your exact Podman/WSL2 error text if you're still stuck. Next class: on to the discrete mathematics and critical thinking content.